Cambridge (CIE) A Level Computer Science

Your notes

Language Translators

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Translator Types

Your notes

What is a translator?

- A translator is a program that **translates** program **source code** into **machine code** so that it can executed directly by a processor
- **Low-level languages** such as **assembly code** are translated using an **assembler**
- **High-level languages** such as **Python** are translated using a **compiler** or **interpreter**

Assembler, compiler, interpreter

What is an assembler?

- An assembler translates **mnemonics written in assembly language** (low-level) in **to machine code**
- **Each lime** of assembly language is **assembled** into a **single machine code** instruction
- Assemblers have been used less and less since high-level languages were introduced

Advantages	Disadvantages
Speed of execution	Difficult to write due to limited and hard to understand commands
Optimises the code	Changes mean it must be reassembled
Original source code will not be seen	Designed solely for one specific processor

What is a compiler?

- A compiler translates high-level languages into machine code **all in one go**
- Compilers translate inputs into machine code **directly**
- Compilers are generally used when **a program is finished** and has been checked for syntax errors
- Compiled code can be **distributed (creates an executable)** and run **without the need for translation software**
- If compiled code contains any errors, after fixing, it **will need re-compiling**

Advantages	Disadvantages
Speed of execution	Can be memory intensive

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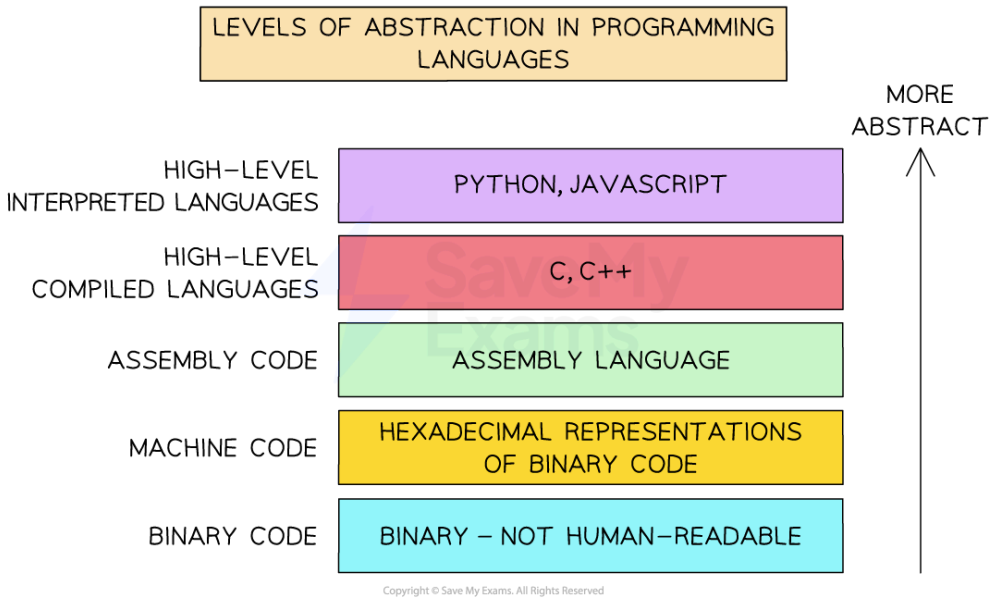
Optimises the code	Difficult to debug
Original source code will not be seen	Changes mean it must be recompiled
	It is designed solely for one specific processor

Your notes

What is an interpreter?

- An interpreter translates high-level languages into machine code **one line at a time**
- Each line is executed after translation and if **any errors are found**, the **process stops**
- Interpreters **do not** generate machine code **directly**, appropriate machine code subroutines are called
- Interpreters are generally used when a program **is being written** in the development stage
- Interpreted code is more **difficult to distribute** as **translation software is needed for it to run**

Advantages	**Disadvantages**
Stops when it finds a specific **syntax error** in the code	Slower execution
Easier to debug	Every time the program is run it has to be translated
Require less RAM to process the code	Executed as is, no optimisation



Compiler vs interpreter

Aspect	Compiler	Interpreter
Software needed by user	End users only need the **executable file**, not the compiler – so there's **no extra cost or setup**	End users **need an interpreter** or compiler to translate and run the source code – may involve setup or cost
Access to source code	Users **do not receive the source code**, so they **can't modify or extend** the program	Users receive the **full source code**, making it **easier to modify or extend** the program themselves
Control and monetisation	Developers keep control of the code and can **charge for upgrades** or customisations	Developers lose control over their code, making it **harder to monetise upgrades** or prevent copying
Speed of execution	Compiled programs **run faster**, as translation is done in advance and optimised machine code is used	Interpreted programs **run slower**, since each line is translated every time it runs
Error handling	Compiled programs don't run until **all syntax and semantic errors** are fixed	Errors are **caught one line at a time**, making it easier for the developer to test and fix during development
Portability	Compiled code can be **run on different machines**, regardless of where it was compiled	Interpreted programs must be interpreted on the **same type of machine** – less portable
Debugging experience	More difficult to test sections – developers need to **write special routines** to check partial results	Developers can **view partial results as they go**, making it easier to test, debug, and refine ideas
Crash risk	Untested compiled programs may **crash the system** if run with errors	Interpreted programs are **safer during testing** – errors are caught before the system can crash
User independence	Users are **reliant on the developer** for updates or modifications – they can't edit the code themselves	Users have **full access to code and libraries**, giving them **freedom to customise or fix** the program

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Mixed mode translation

What is mixed mode translation?

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- Mixed mode translation **combines features** of both a **compiler** and an **interpreter**
- Instead of fully compiling or interpreting the code, the program is **partially compiled into an intermediate form**, which is then **interpreted at runtime**
- It works by:
  - The source code is **compiled into intermediate code**, not full machine code
  - This **intermediate code** (e.g. bytecode) is **interpreted by a virtual machine** (e.g. the Java Virtual Machine – JVM)
  - Optionally, some parts may later be **just-in-time (JIT) compiled** into machine code for better performance
- **Java** uses mixed mode translation:
  - Java source code is **compiled** to **bytecode** (.class file)
  - Bytecode is **interpreted** (or JIT compiled) by the **JVM** on any platform

Feature	Compiler	Interpreter	Mixed mode translation
Translation	Entire program at once	One line at a time	Compiles to intermediate code
Speed of execution	Fastest (fully compiled)	Slowest	Medium (JIT makes it faster over time)
Portability	Not portable	Not portable	Highly portable (across platforms)
Errors shown	After full compilation	One at a time	During compile or runtime
Requires runtime system	No	Yes	Yes (e.g. JVM)
Example languages	C, C++	Python, JavaScript	Java, C# (with .NET CLR)

Worked Example

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Jennifer uses an Integrated Development Environment (IDE) to write her computer program.

The IDE allows Jennifer to use both an interpreter and a compiler while creating her computer program.

Describe the ways in which Jennifer can use both a compiler and an interpreter while developing the program. **[4]**

Answer

Interpreter:

- Use an interpreter while writing the program [1 mark]
- … to test/debug the partially completed program [1 mark]
- … because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time [1 mark]

Compiler:

- Use the compiler after the program is complete [1 mark]
- … to create an executable file [1 mark]
- Use the compiler to repeatedly test the same (completed) section [1 mark]
- … without having to re-interpret every time // compiler not needed at runtime [1 mark]

Your notes

Your notes

What is an IDE?

- An **Integrated Development Environment (IDE)** is a software tool that provides programmers with a comprehensive and integrated platform to **write**, **edit**, **compile**, **debug**, and **manage** their code efficiently
- IDEs are designed to **streamline the software development process** by offering a wide range of features and tools that can help programmers
- IDEs can be an **app to download** or can be **web-based**

IDE features

Syntax highlighting (pretty printing)

- IDEs use syntax highlighting to **visually distinguish** different elements of code based on their syntax
- The IDE assigns **distinct colours or styles** to keywords, strings, comments, variables, functions, and other language constructs
- Syntax highlighting helps programmers **quickly identify and differentiate code** components, reducing the chance of syntax errors and improving code comprehension
- In this example Python code, you can see that the code comments are displayed in red, the strings are displayed in green and the print commands are displayed in purple

```
# global variable
global_var = "I am a global variable"

def modify_global_var():
    # declaring variable as global
    global global_var
    # modifying global variable
    global_var = "I have been modified"

print(global_var)  # outputs: I am a global variable
modify_global_var()
print(global_var)  # outputs: I have been modified
```

Syntax highlighting in an IDE

Autocomplete

- Autocomplete will **suggest code completions** as programmers type
- For example, as developers start typing a variable, function, or method name, the IDE displays a list of available options

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- Another example is when programmers use opening brackets and the autocomplete facility will automatically add closing brackets
- Code completion **improves productivity** by reducing typing effort and preventing typing errors and naming inconsistencies
- IDEs can automatically correct common mistakes or typing errors made during coding. When an error-prone sequence is detected, **the IDE can suggest a correction**, helping developers fix minor errors as they type their code

```
13
14  modify_
    f modify_global_var        def modify_global_var() -> None
```

Code completion in an IDE

Auto indent

- Auto indent will **automatically indent the programming code** when you start a new line of code within a code block
- They are commonly used in **selection** and **iteration** programming constructs
- In the example below, the indentation is automatic after starting to define the code block on line 04 and is clearly marked to show which lines of code are included within the block.

```
4  def modify_global_var():
5      # declaring variable as global
6      global global_var
7      # modifying global variable
8      global_var = "I have been modified"
9
```

Code formatting with automatic indentation

Code comments

- Comments are **lines of text within the code that are not executed** but serve as documentation and explanations for programmers
- IDEs enable developers to add single-line or multi-line **comments to describe the code's purpose**, algorithmic steps, and any relevant information that will be useful to the maintenance of the program
- Comments enhance code understanding, making it easier for developers to revisit and modify code at a later stage
- Comments are in green on the image above

Syntax error highlighting

Your notes

- IDEs can automatically highlight **syntax errors in the code editor** as developers type
- This feature **helps catch mistakes instantly**, allowing programmers to address syntax-related issues promptly
- Syntax errors are highlighted with distinct colours, making them easily noticeable, even before executing the code

Your notes

Variable watch window

- The watch window is a debugging tool that allows developers to **monitor the values of specific variables** during **runtime**
- Developers can add variables to the watch window and observe their values change as the program executes
- This tool is particularly valuable for understanding how variables evolve throughout the program's execution and diagnosing issues related to variable values

Breakpoints

- Breakpoints are **markers** that developers can set at specific lines of code to **halt program execution** during **runtime**
- When the program reaches a breakpoint, it pauses, allowing developers to inspect the program's state and variable values and identify the cause of potential bugs
- Breakpoints provide a controlled way to analyse code step-by-step

Error message list

- The error message list **displays a collection of errors**, **warnings**, and **messages** generated during the compilation and execution of the program
- IDEs present detailed error descriptions and relevant code locations, enabling programmers to address and correct issues systematically

Stepping mode

- Stepping mode is a debugging mode that allows developers to **execute the program one line at a time**
- Developers can choose between different step options, such as step into, step over, and step out, to examine function calls and control flow
- Step mode offers a granular view of the program's execution, making it easier to identify the exact location of potential bugs

Crash dump/post-mortem report

- When a program crashes, IDEs can **generate** a crash dump or post-mortem report
- This report **captures the program's state** and **memory** contents **at the time of the crash**, helping developers diagnose the root cause of the crash after the program has terminated unexpectedly

Stack contents

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- The stack contents tool allows developers to **inspect the contents of the program's call stack** during debugging
- It shows the order in which functions were called and the parameters and local variables of each function
- Understanding the call stack, aids in understanding the program's flow and diagnosing issues related to function calls

Your notes

Worked Example

Identify two debugging tools that a typical IDE can provide. **[2]**

Answer

e.g.

- Breakpoints [1 mark]
- Single stepping [1 mark]
- Report windows [1 mark]

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